TEAM NUMBER: 65

SwaraAI Progress report: what works, what broke, how we fix it

Voice-first dining assistant for blind, low-vision, deaf and mute diners in Indian restaurants. Status as of 11 Oct 2026, 03:30 IST.

1 Progress at a Glance

STATUS

SvaraAI is a native Android app (Kotlin, Jetpack Compose, Room) that lets blind, low-vision, deaf and mute diners order, talk to staff and pay in Indian restaurants. **The speech, ordering, waiter-communication and billing engines are working end to end. Menu capture from the camera is the one component still being repaired**: it fails on real restaurant menus, and the root causes are now diagnosed. This report covers the approach as built, the experiments that exposed the failures, and our testing progress.

Capability	Status	What is built (evidence in code)
Bhashini speech and TTS	Working	ASR (Conformer), IndicTrans2 NMT, Indic TTS; 16 kHz PCM capture with silence thresholding; 11 languages
Phonetic dish normalizer	Working	**DishNormalizer**: Levenshtein plus soundex over 150+ dishes; quantity words and diet modifiers extracted
Dual-sensory feedback	Working	**HapticFeedbackEngine** (6 patterns), **EarconEngine** (4 tones), OLED-black high-contrast theme, 48 dp targets
Voice ordering and cart	Working	Speech to **CartItemEntity** with cooking notes; Room Flow sync; one-tap spoken cart summary
Waiter Bridge	Working	Outward card rotated 180 degrees, spoken phrases in the waiter's language, quick request chips
Smart bill and GST	Working	**BillCalculator** in integer paise: CGST 2.5% + SGST 2.5%, optional 5% service charge and tip; **BillSpeechNarrator**
Multi-diner splitter	Working	Equal split with remainder paise given to first diners; item-by-item split with proportional tax
Camera menu capture	Broken	CameraX analysis and cloud OCR work in the lab but fail on real menus (Section 5)

2 Approach as Implemented

The M1 architecture held up. What we have actually built follows four rules, and each one has paid off in testing so far:

- **Bhashini as the speech spine.** The same ASR, NMT and TTS services serve ordering, bill narration and the Waiter Bridge, so one integration covers every language.
- **Heavy code-switching is handled before the LLM.** Orders such as "Ek Masala Dosa aur do filter coffee dena" are cleaned by the local DishNormalizer first, so the LLM (Groq LLaMA-3.3-70B) sees clean dish names, counts and modifiers.
- **Deterministic money.** All bill and split maths is **Long** paise, never floating point, so shares always add up to the exact total.
- **Graceful degradation.** If a cloud TTS round trip exceeds 1800 ms the app switches to Android TextToSpeech; common prompts and earcons are cached; network calls return **Result<T>**, so a dropped connection never crashes the app.

Change since M1. The M1 plan described a camera-first journey. Real-menu testing showed that this front door is the weakest part, so we prioritised the voice, waiter and billing paths first and now treat camera capture as a repair task with four targeted fixes instead of a redesign.

3 Code Map: What Is in the Repository

Module	Responsibility
AppDatabase, MenuDao, CartDao, OrderDao	Room persistence for menu, cart and orders; live Kotlin Flow updates to screen, caption bar and bill
AudioRecorder + Bhashini clients	16 kHz PCM capture, silence and energy thresholding, ASR / NMT / TTS requests
DishNormalizer	Levenshtein and soundex matching, quantity words, diet modifiers
BillCalculator, BillSpeechNarrator	Integer-paise GST, service charge and splits; spoken bill in the chosen language
HapticFeedbackEngine, EarconEngine	6 vibration patterns and 4 synthesised tones for state feedback

4 Experiments Run So Far

Experiment	Setup	Result
Code-switched ordering	Spoken orders mixing Hindi and English dish names through ASR then DishNormalizer	Misheard forms are corrected: "do sa" to Dosa, "idly" to Idli, "paneer butter masala" to Paneer Butter Masala
Diet modifiers	Phrases such as "without onion", "jain", "less spicy", "extra butter"	Extracted into the cart note and stored in Room with the item
Bill narration	Cart of 460 rupees, service charge on, spoken in a chosen language	460 + 23 service + 11.50 CGST + 11.50 SGST = **506 rupees**, read out in full
Group split	Total divided among N diners; item-by-item assignment	Shares sum to the exact total in paise; remainder goes to the first diners
Cloud TTS latency	Network round trips on mobile data and weak signal	Over 1800 ms triggers the local TTS fallback so speech never stalls
Real-menu capture	Camera capture on real, laminated, multi-column Indian menus and on a cluttered desk	**Four failures found** (below); upload to cloud OCR and LLM takes 4 to 8 s

5 Failures Found, Root Causes and Fixes

Failure observed	Root cause	Targeted fix (no redesign)	Status
Keyboards, tabs and cutlery trigger capture	Edge-contrast check only; no document shape check	**Fix 1**: four-corner contour validation in CameraX before capture	Next
Multi-column text merges; headers lose their dishes	OCR returns a flat string; 2D coordinates are dropped	**Fix 2**: keep (X, Y) per word block so columns stay separate	Next
Prices attach to the wrong item (drink price on a main)	LLM gets flat text with no line anchors	**Fix 3**: row-anchored prompt; strict line-by-line price pairing	Next
Audio freezes for 4 to 8 s after capture	Whole cloud chain finishes before any speech starts	**Fix 4**: play local earcon at once, stream category names as parsed	Next

Acceptance criteria for re-testing the camera pipeline (same menus and same desk scene as the failed runs):

Check	Pass condition
False triggers	No capture on keyboard, browser tab, table or cutlery; capture only on a detected document
Structure	Every category header stays with its own dishes on multi-column menus
Price pairing	Each price attaches to the correct dish on every test menu
Responsiveness	Speech cue starts immediately on capture; no multi-second silence while parsing continues

6 Testing Progress and Plan to Final

Test area	Done so far	Still to do before the 11:00 AM final
Voice and language	Ordering, cart notes and narration exercised in the working languages	Noise test in a loud room; native-speaker check of Hindi and Telugu phrasing
Billing and splits	Worked examples (506 rupees; exact-sum splits) reproduce correctly	Edge cases: one-paisa remainders, tip with service charge off, single diner
Accessibility	Six haptic patterns, four earcons, high-contrast theme in place	TalkBack walkthrough of every screen; touch-target and contrast scan
Camera capture	Failures reproduced and root-caused on real menus	Re-run on the same menus after Fixes 1 to 4; compare false triggers, item and price accuracy, time to first speech
Resilience	Cloud-to-local TTS switch at 1800 ms; Result<T> error handling on all network calls	Airplane-mode run of ordering, cart and bill; injected timeouts and dropped connections

Priorities to the final. (1) Ship Fixes 1 to 4 and re-test on the same real menus. (2) Complete the remaining test passes in the table above. (3) Prepare the final package: README with run instructions, LICENSE, 3-page final report and demo video. **Code push:** all scripts and modules built so far are committed to the team repository on **main**; third-party services (Bhashini, Groq) are credited in the README.